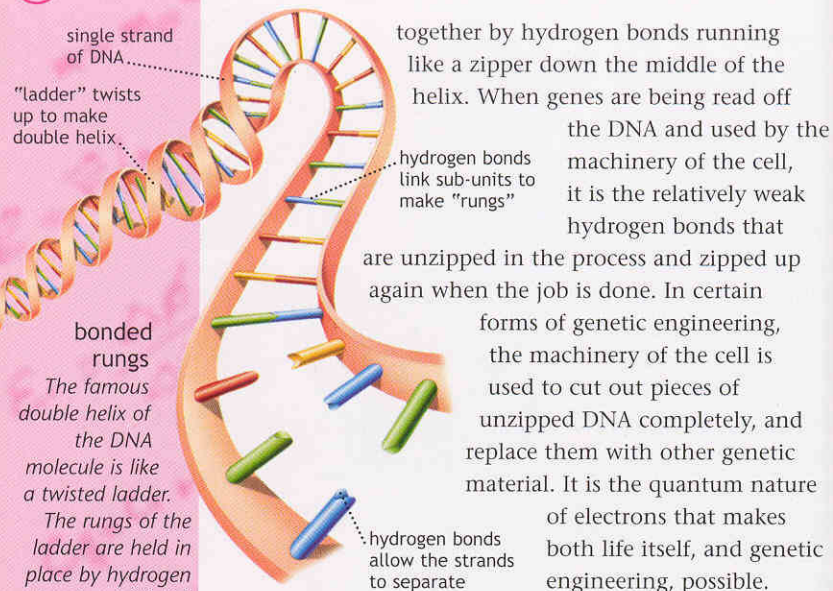
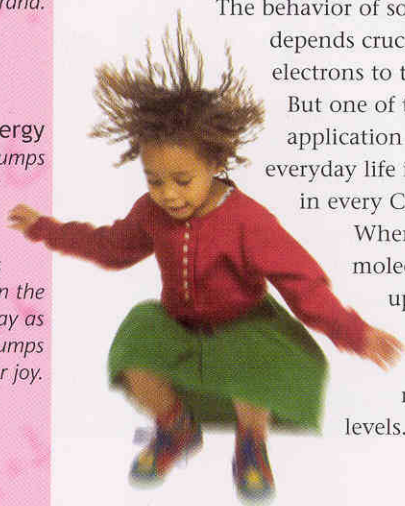




bonded rungs
The famous double helix of the DNA molecule is like a twisted ladder. The rungs of the ladder are held in place by hydrogen bonds, which link the half-rung from one strand of the molecule to the half-rung on the other strand.

high energy
An electron jumps up to a higher energy level in an atom when it is excited, in the same way as a child jumps for joy.



together by hydrogen bonds running like a zipper down the middle of the helix. When genes are being read off the DNA and used by the machinery of the cell, it is the relatively weak hydrogen bonds that are unzipped in the process and zipped up again when the job is done. In certain forms of genetic engineering, the machinery of the cell is used to cut out pieces of unzipped DNA completely, and replace them with other genetic material. It is the quantum nature of electrons that makes both life itself, and genetic engineering, possible.

laser technology

The quantum properties of electrons are also important in electronics, and especially in the design of microchips.

The behavior of some circuits, for example, depends crucially on the ability of electrons to tunnel through barriers. But one of the neatest examples of the application of quantum physics in everyday life is the laser, a device found in every CD player.

When a collection of atoms or molecules gets hot, they pick up energy (they are said to be "excited"), and the electrons in them are raised up to higher energy levels. Left to cool off, these

electrons jump back down to lower energy levels radiating light in a more or less random fashion to produce the black body curve (see page 14). But if a weak input of radiation with just the right energy is fed into the right kind of material, the effect is to raise the electrons in each atom (or at least, very many atoms) of the material up into the same excited state as all their neighbors. Now, when they jump back down from the excited state, they will each emit a photon with exactly the same amount of energy—which means precisely the same wavelength. It is all these billions of photons marching precisely in step with each other that form the intense beam of pure monochromatic light of a laser (the name comes from Light Amplification by Stimulated Emission of Radiation).

The principle is very simple—it actually goes back to some calculations made by Einstein in 1916. But the technological difficulty of getting a lot of atoms into the same excited state and keeping them there until you are ready to trigger the release of energy proved so great that the first lasers were not developed until 40 years after Einstein's calculations. Now, you can get one given away free with a sound system, to hang on your keychain. There is no better example of how far quantum physics has become part of everyday life since the days of Einstein and Bohr. But there are still aspects of the quantum world to be explored and turned into new technologies for the future. We discuss some of these ideas in the next section.



laser display
The first public performance of a laser show took place on May 9, 1969, at Mills College in Oakland, California. Today, the laser beams are made to move by mirrors controlled by magnets and electronics that can move the beam faster than the eye can follow.

miniature version
A tiny pocket laser attached to a keychain is an example of quantum technology, based on an idea from Einstein that is now commonplace.

